

## MTH209: Assignment 12

### Prediction for LDA

Recall that the discriminant function for a class  $k = 1, 2, \dots, K$  in LDA is

$$\delta_k(x) = \log \pi_k + \mu_k^\top \Sigma^{-1} x - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k$$

Consider the problem of predicting the class label given all parameters. Create a function `predict_lda` that takes appropriate inputs and outputs 1, 2, ..., K based on the predicted class label.

```
# x is a vector. Note, this must be a vector and not a matrix.
# mus: a list of K vectors
# Sigma: the common covariance
# pis: probabilities of all populations. Assumed to sum to 1.
predict_lda <- function(x, mus, pis, Sigma)
{
  K <- length(mus)
  # mus[[1]] is the first list element
  ...
  ...
  label <- # either 1, 2, ..., K (numeric)
  return(label)
}
```

In case it is helpful for testing, to make a list in R test using the following

```
K <- 3
mus <- list()
mus[[1]] <- c(3,5)
mus[[2]] <- c(-4, -8)
mus[[3]] <- c(-2, 8)
mus
mus[[2]]
```

**NOTE:** Only upload the function `predict_lda` in `assignment12.R` and nothing else.